

ASTRA INTEL | Unclassified // Training sample
C2 Integration Whitepaper for Joint UAS Employment

This paper describes how unmanned systems plug into a joint command and control architecture. It complements the 2024 UAS capability brief.

Coalition tasking depends on common datalink profiles and shared track numbering so that a NATO Combined Air Operations Centre can reassign an air vehicle without a full re-brief of the aircrew-equivalent mission team on the ground.

Interoperability notes

STANAG 4586 remains the baseline control interface. National extensions must not break the core message set used for take-off, payload cueing and lost-link handling.

Compared with platform-centric briefs, this paper emphasises the ground segment: multi-vehicle control, spectrum deconfliction, and the human authorisation loop for any munition release.

Differences from platform capability documents

Capability briefs list sensors and air vehicle classes. This whitepaper instead lists C2 failure modes: SATCOM fade, control-station power loss, and cross-domain guard delays when tracks move to classified networks.

Similarities include the same requirement for redundant C2 links and the prohibition on autonomous target engagement without a human in the loop.

Recommendations

Fund dual-path datalinks before additional airframes.

Exercise lost-link and GNSS-denied profiles quarterly.

Keep a common operational picture feed into the maritime operations centre for ship-to-shore surveillance missions.